S4 Supplementary Clinical Summary

A brief, cautious clinician-facing note on this mechanistic/methods-level finding

Article type: Supplement

Short title: S4 Clinical Summary - RichSleep_EAP_v2

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Important: not yet a clinical finding

This article reports a basic-science, mechanistic/methods-level survey of how brain rhythms during sleep relate to breathing and heartbeat timing. It tested **no** clinical population, intervention, or outcome. The single dataset (RichSleep, 24 subjects) is healthy and unselected. Nothing here supports any diagnostic, monitoring, or intervention use.

What was found, in plain language

Slow brain waves that occur during the deepest stage of sleep (slow oscillations) line up strongly and consistently with a particular point in the breathing and heartbeat cycle - more strongly than anything else measured in this research program so far, across 24 healthy adults. A memory-related rhythm that occurs during dream sleep (REM theta) showed no such link at all. For the sleep spindle - a shorter burst of brain activity in lighter sleep, studied in a companion article - three different, individually reasonable ways of testing whether its very **start** lines up with breathing/heartbeat gave three different answers; rather than picking the one we prefer, we report the disagreement itself as the finding. Finally, a separate check asked whether a spindle's own intrinsic strength changes how much a preferred-timing effect (found in the companion study) shows up in the brain's state-space geometry: it does not, at least not detectably with the data available.

Why this is not (yet) clinically actionable

- This is a **timing** finding about when different sleep rhythms align with breathing/heartbeat - not a finding about memory, cognition, sleep quality, or any clinical outcome. None of those were measured or tested here.
- The single dataset is healthy, unselected, and observational; no sleep disorder, cognitive condition, or intervention was studied.
- The one question with the most attempted rigor - whether spindle onset timing predicts anything - produced three disagreeing answers rather than one, which is reported honestly as unresolved rather than as a clean result to build on.
- The strength-moderation check (does a stronger spindle show a bigger timing-related effect on brain-state geometry) was clean-null across three different ways of measuring "strength" - explicitly not evidence that intrinsic strength matters, nor that it doesn't; only that no such interaction was detected at current sample size.
- "Statistically significant" in this article, as elsewhere in this research program, is a multiple-comparison-corrected statistical statement, not a claim about clinical meaningfulness or effect size at the individual level.

What would make this more clinically relevant

- Replicating the slow-oscillation timing result in an independent, ideally larger and/or clinical (e.g. sleep-apnea, insomnia, dementia-risk) PSG sample - this is the strongest and most replication-worthy result in this article.
- A substantially larger slow-oscillation-spindle-coupled event yield (this article's coupled subset, 25 events/15 subjects, was too small for a confirmatory test), to properly power a genuine strength-moderation test.
- Resolving, rather than merely reporting, the three-way disagreement on spindle-onset timing - e.g. with a purpose-built, pre-registered null construction chosen in advance of seeing any result.
- Any demonstration that carrier-specific autonomic phase-locking, or its downstream geometric consequence, differs between healthy and clinical populations - entirely untested here.

Summary for clinical readers

This is a basic-science survey of **which** sleep rhythms show timing links to breathing/heartbeat (strongest for slow oscillations, essentially absent for REM theta, and unresolved for sleep-spindle onset), together with a clean-null check of whether spindle strength moderates a companion study's more tentative timing-geometry

link. None of this currently supports any diagnostic, monitoring, or intervention use; the slow-oscillation result is the most clinically-promising thread for future replication work.

Bibliography